

Electric Vehicle Adoption Analysis in India

Power BI Project Presentation

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Project Introduction

The objective of this project is to analyze Electric Vehicle (EV) adoption trends across India.

Key Goals:

- Analyze EV sales growth
- Identify top-performing states
- Compare vehicle categories
- Understand EV adoption patterns
- Generate actionable insights

Dataset Overview

Source

Dataset collected from Kaggle EV adoption and registration data across Indian States & UTs
Analysis period: 2014 – 2023

Dataset Contains:

- State Name
- Vehicle Category
- Vehicle Type
- EV Sales
- Year

Data Cleaning & Transformation

Data Preparation

- Removed duplicate records
- Checked and corrected data types
- Removed unnecessary column
- Created Date Table
- Prepared data for reporting

Tools Used: Power Query Editor

Data Model

Model Type: Star Schema

Tables Used:

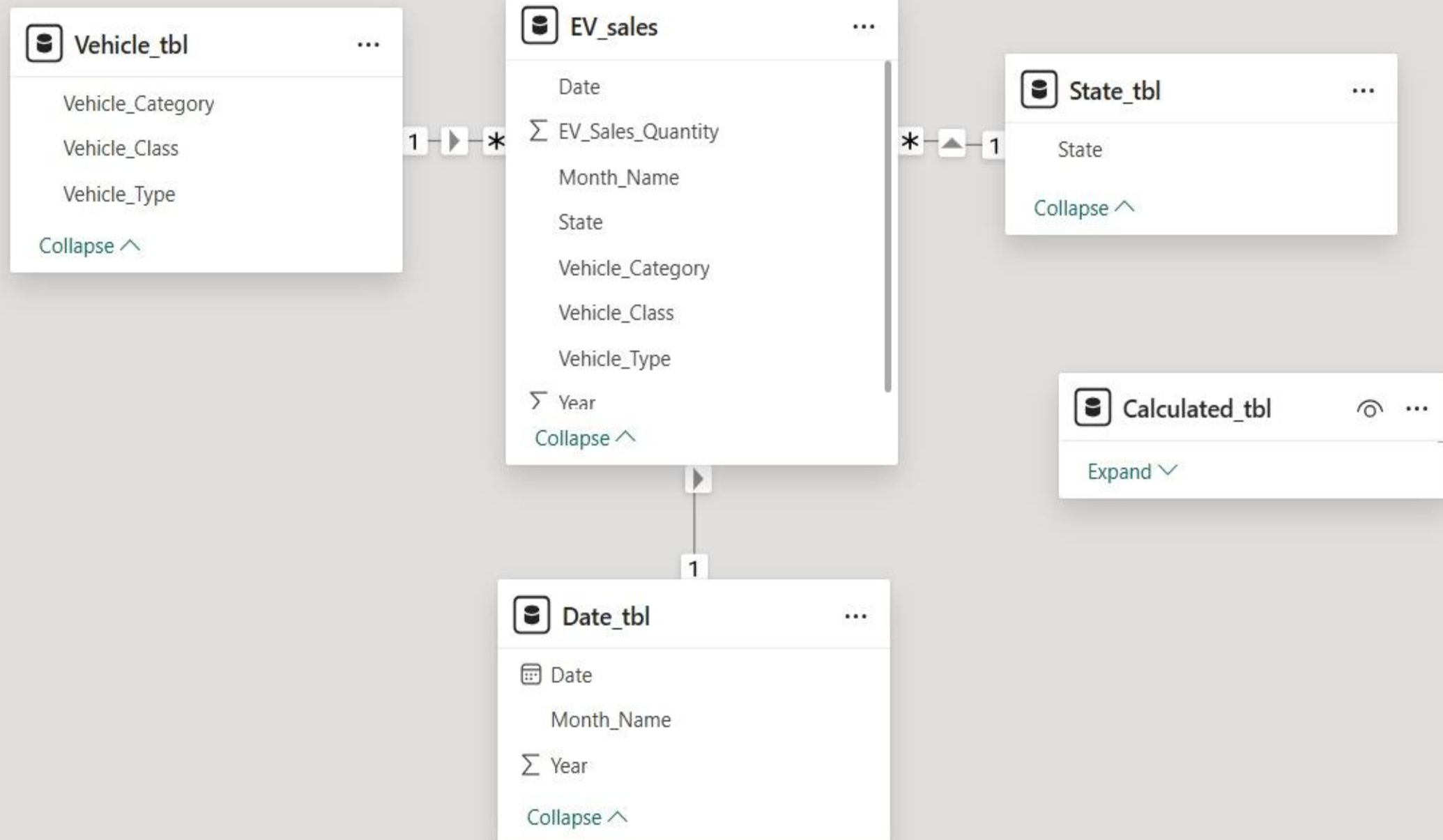
Fact Table: EV Sales

Dimension Tables: Date Table, State Table, Vehicle Table

Relationship type (One to Many)

Benefits:

- Improved performance
- Easy filtering
- Better report design

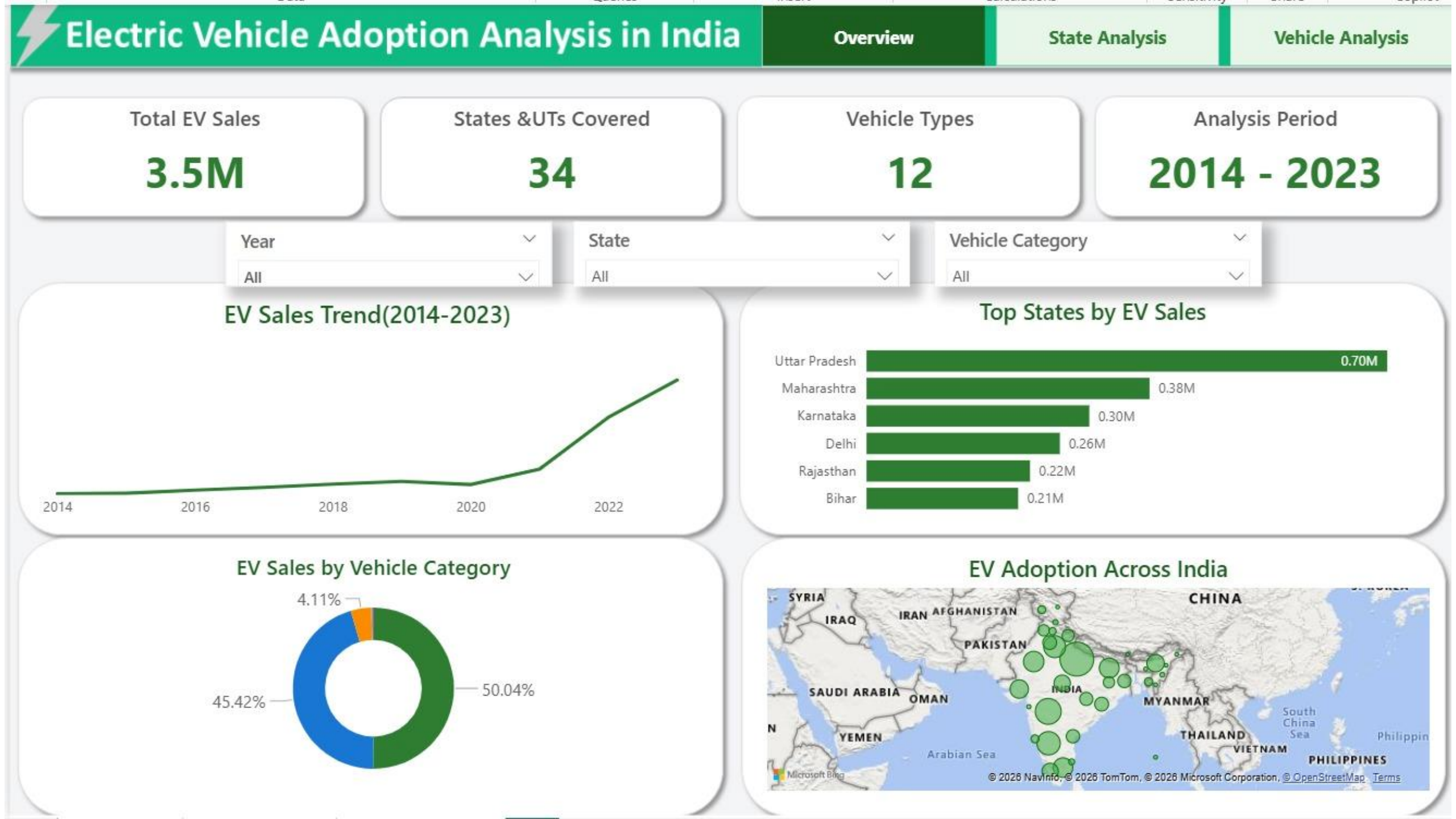


DAX Measures

Measures Created:

- Total EV Sales
- Total States
- Total Vehicle Types
- Top Performing State
- Lowest Performing State
- Top State Contribution %
- Top Vehicle Contribution %

Dashboard Overview



Top Performing State

Uttar Pradesh

Lowest Performing State

Sikkim

State Contribution

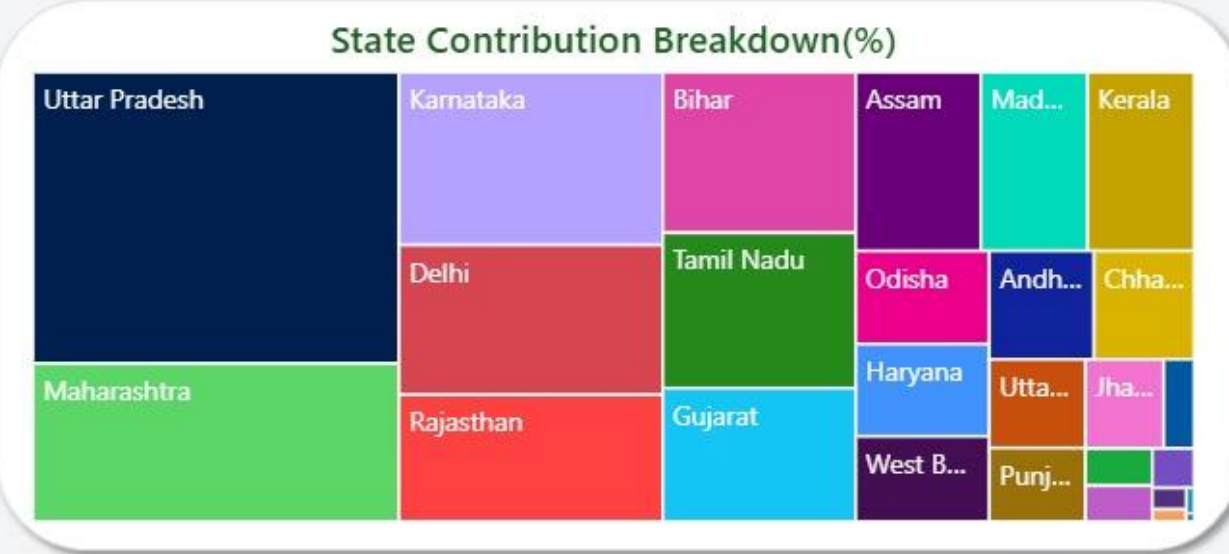
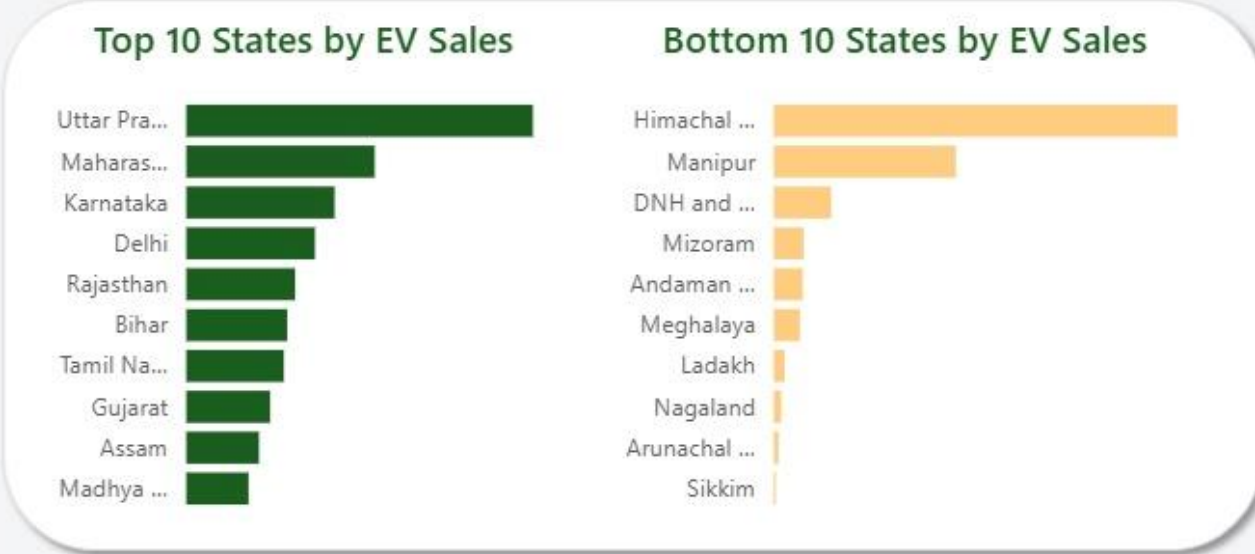
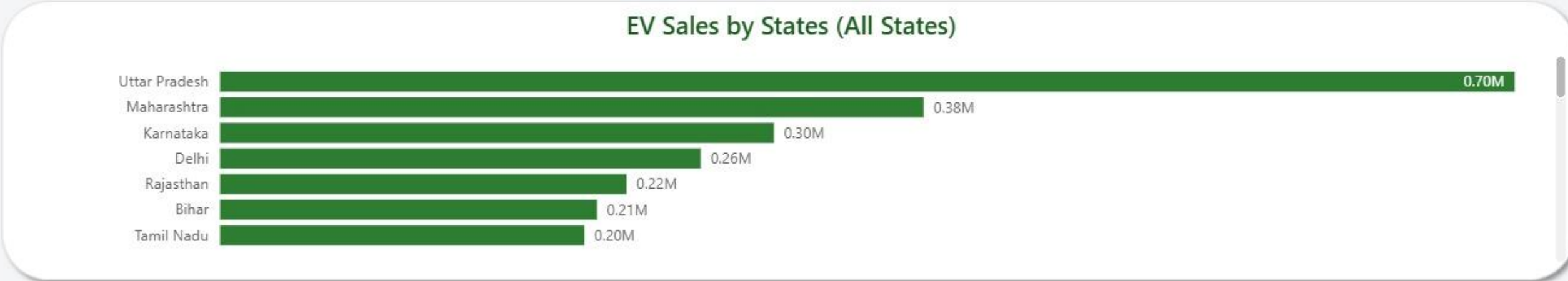
20.4%

Year

All

State

All



Top Vehicle Category

2-Wheelers

Lowest Vehicle Category

Bus

Vehicle Contribution

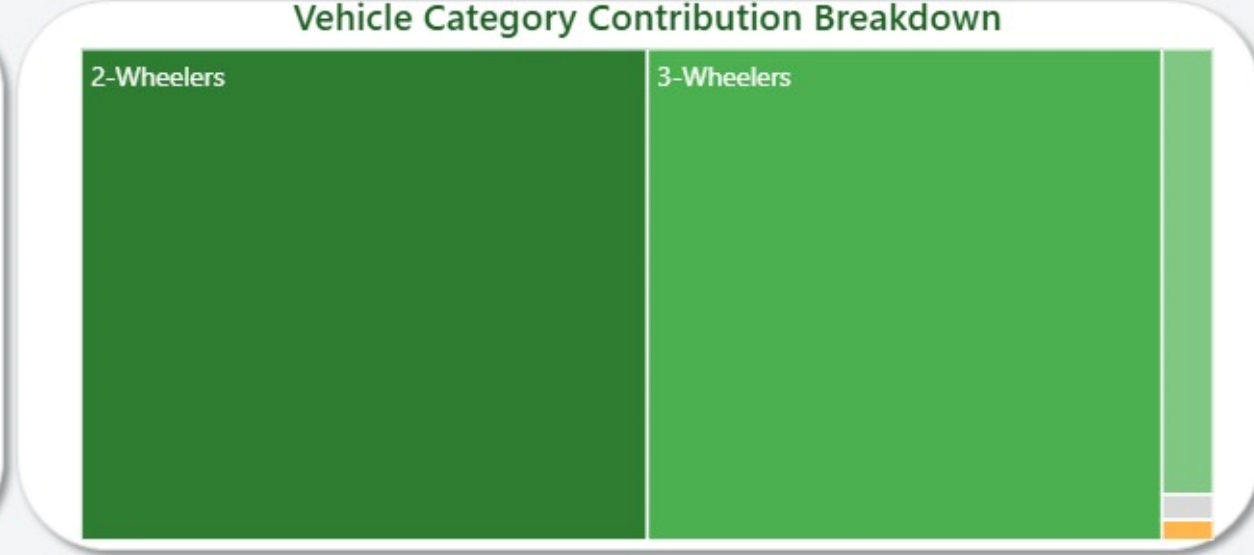
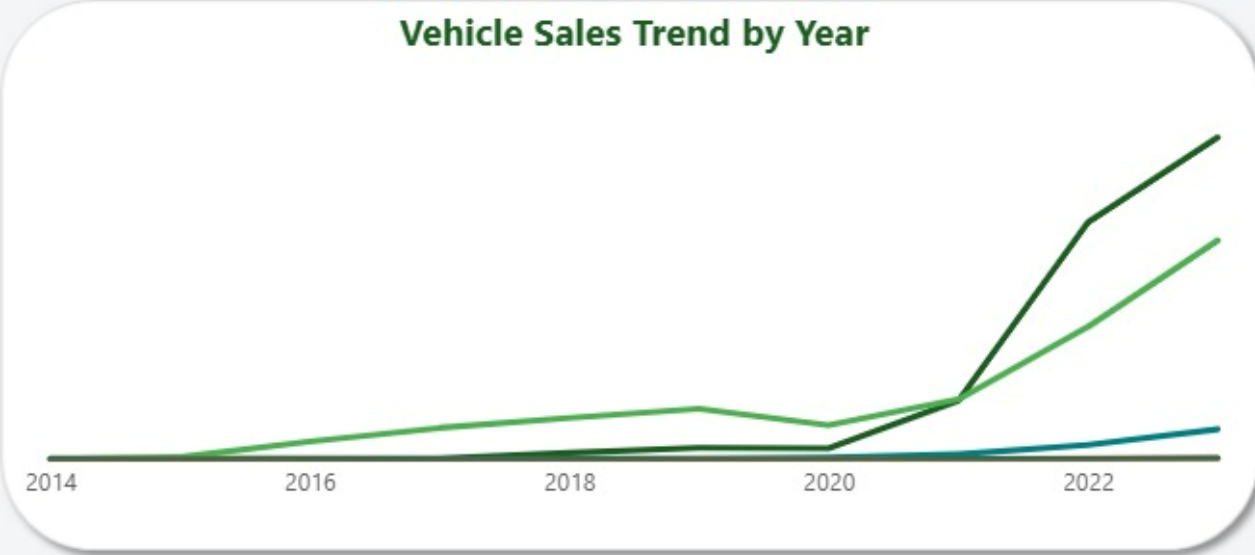
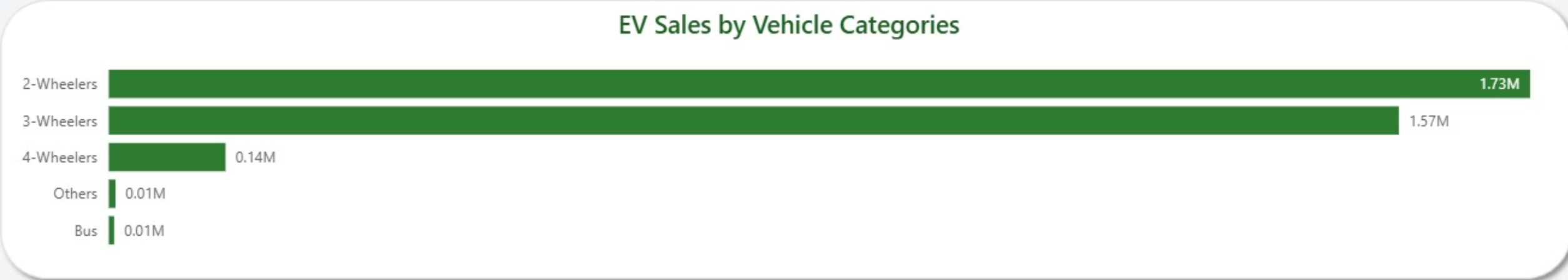
50.0%

Vehicle Type

All

Vehicle Categ...

All



Key Insights

Major Findings

- Total EV Sales exceeded 3.5 Million.
- Uttar Pradesh is the Top-performing state.
- Sikkim has the lowest EV sales.
- 2-Wheelers dominate the EV market.
- EV adoption increased significantly after 2021.

Future Scope

- EV Sales Forecasting
- Real-Time Dashboard Integration
- Regional Demand Prediction
- Charging Infrastructure Analysis
- Government Policy Impact Analysis

Conclusion

The analysis shows a strong increase in EV adoption across India.

- Strong growth in EV adoption
- 2-Wheelers dominate the EV market
- Significant variation across states
- Dashboard supports data-driven decisions

Thank You